

Angular Momentum Representation For Free particle

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Abstract

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Keywords: a

1 Problem Setting

The purpose of this project is to construct a complete and self-contained formulation of the angular-momentum representation for a nonrelativistic free particle. The central objective is to solve the time-dependent Schrödinger equation (TDSE)

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle, \quad \hat{H} = \frac{\hat{\mathbf{p}}^2}{2m},$$

in a basis that diagonalizes simultaneously the free Hamiltonian $\hat{H}$ and the angular-momentum operators $\hat{L}^2$ and $\hat{L}_z$.

The motivation for this construction is threefold:

1. To reveal the role of rotational symmetry in quantum mechanics through the generators of the group $\text{SO}(3)$.
2. To establish a clear connection between spatial symmetries and conservation laws.
3. To demonstrate how angular-momentum eigenstates provide a natural separation of variables for free-particle dynamics in spherical coordinates.

1.1 Physical Scenario

We consider a single spinless particle of mass m moving freely in three-dimensional space. No external forces act on the system, so the Hamiltonian contains only kinetic energy:

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m}.$$

Because the dynamics is invariant under translations and rotations, the corresponding generators $\hat{\mathbf{p}}$ and $\hat{\mathbf{L}}$ play a central role. In particular:

$$[\hat{H}, \hat{\mathbf{L}}] = 0,$$

which implies that $\hat{H}$, $\hat{L}^2$ and $\hat{L}_z$ admit simultaneous eigenstates.

1.2 Goal of the Construction

We aim to:

- Construct the operators $\hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}}$ from the canonical commutation relations.
- Express $\hat{\mathbf{L}}$ in both Cartesian and spherical coordinates.
- Establish the commutation relations that define the Lie algebra of $\text{SO}(3)$:

$$[\hat{L}_i, \hat{L}_j] = i\hbar \epsilon_{ijk} \hat{L}_k.$$

- Solve the eigenvalue problems for $\hat{L}_z$ and $\hat{L}^2$, obtaining the spherical harmonics $Y_{\ell m}(\theta, \phi)$.
- Expand the free-particle wavefunction in the basis $|k, \ell, m\rangle$.
- Solve the TDSE in this basis and reconstruct the wavefunction in position, momentum, and energy representations.

1.3 Mathematical Background Assumed

The development assumes familiarity with:

- Hilbert-space formalism: kets, operators, inner products.
- Canonical commutators $[\hat{x}_i, \hat{p}_j] = i\hbar \delta_{ij}$.
- Fourier transforms and the relation between position and momentum representations.
- Basic knowledge of spherical coordinates and special functions (Legendre polynomials).

1.4 Expected Outcome

The final result of this construction is the full decomposition

$$|\psi(t)\rangle = \sum_{\ell, m} \int_0^\infty dk k^2 \psi_{k\ell m}(0) e^{-i\hbar k^2 t/(2m)} |k, \ell, m\rangle,$$

along with the corresponding integral transforms that yield the wavefunction in position-, momentum-, and energy-space representations. This framework encapsulates the full rotational structure of the free particle and sets the stage for generalizations to interacting systems, spin, and relativistic dynamics.

2 Starting Point: Known Results

In this section, we summarize the fundamental elements of the quantum mechanical formalism for a free particle of mass m . These results are assumed to be known and serve as the foundation for constructing the angular momentum representation. Proofs are omitted; only the necessary definitions and operational relations are presented.

2.1 Free Particle Solution in Different Representations

The free particle Hamiltonian is given by $\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m}$. The wavefunctions in the various representations evolve as follows:

- **Position Representation** ($\psi(\mathbf{r}, t)$): Satisfies the time-dependent Schrödinger equation:

$$i\hbar \frac{\partial}{\partial t} \psi(\mathbf{r}, t) = -\frac{\hbar^2}{2m} \nabla^2 \psi(\mathbf{r}, t). \quad (1)$$

- **Momentum Representation** ($\psi(\mathbf{p}, t)$): Since $\hat{\mathbf{p}}$ commutes with $\hat{H}$ for the free particle, the evolution is a multiplicative phase:

$$\psi(\mathbf{p}, t) = \psi(\mathbf{p}, 0) \exp\left(-\frac{i}{\hbar} \frac{p^2}{2m} t\right). \quad (2)$$

- **Energy Representation** ($\psi(E, t)$): For an energy eigenstate with $E = p^2/2m$:

$$\psi(E, t) = \psi(E, 0) \exp\left(-\frac{i}{\hbar} E t\right). \quad (3)$$

2.2 Integral Transformations Between Representations

The connection between different bases is established through integral transformations defined by the inner products (kernels) of the basis states.

2.2.1 Fourier Transform ($\mathbf{r} \leftrightarrow \mathbf{p}$)

The wavefunctions are related by:

$$\psi(\mathbf{r}, t) = \frac{1}{(2\pi\hbar)^{3/2}} \int d^3p e^{i\mathbf{p}\cdot\mathbf{r}/\hbar} \psi(\mathbf{p}, t), \quad (4)$$

$$\psi(\mathbf{p}, t) = \frac{1}{(2\pi\hbar)^{3/2}} \int d^3r e^{-i\mathbf{p}\cdot\mathbf{r}/\hbar} \psi(\mathbf{r}, t). \quad (5)$$

2.2.2 Momentum-Energy Transformation ($\mathbf{p} \leftrightarrow E$)

Using the dispersion relation $E = p^2/2m$ and the properties of the deformed Dirac delta:

$$\delta\left(E - \frac{p^2}{2m}\right) = \frac{m}{p} \delta(p - \sqrt{2mE}). \quad (6)$$

This allows expressing integrals over momentum as integrals over energy, facilitating the transition from $\psi(\mathbf{p})$ to spectral energy densities.

2.3 Eigenvalue Equations and Basic Operators

We define the fundamental operators and their eigenstates, which form complete and orthonormal bases.

- **Position Operator $\hat{\mathbf{r}}$:**

$$\hat{\mathbf{r}}|\mathbf{r}\rangle = \mathbf{r}|\mathbf{r}\rangle, \quad \int d^3r |\mathbf{r}\rangle \langle \mathbf{r}| = \hat{\mathbb{I}}. \quad (7)$$

- **Momentum Operator $\hat{\mathbf{p}}$:**

$$\hat{\mathbf{p}}|\mathbf{p}\rangle = \mathbf{p}|\mathbf{p}\rangle, \quad \int d^3p |\mathbf{p}\rangle \langle \mathbf{p}| = \hat{\mathbb{I}}. \quad (8)$$

In the position basis, the momentum operator acts as the generator of translations:
 $\hat{\mathbf{p}} = -i\hbar \nabla_{\mathbf{r}}$.

- **Free Hamiltonian $\hat{H}$:** Its eigenstates $|E\rangle$ coincide (with degeneracy) with the momentum states:

$$\hat{H}|\mathbf{p}\rangle = \frac{p^2}{2m}|\mathbf{p}\rangle. \quad (9)$$

- **Inner Products (Transformation Kernels):**

$$\langle \mathbf{r}|\mathbf{p}\rangle = \frac{1}{(2\pi\hbar)^{3/2}} e^{i\mathbf{p}\cdot\mathbf{r}/\hbar}, \quad \langle \mathbf{p}|E\rangle \propto \delta(E - p^2/2m). \quad (10)$$

2.4 Symmetry Operators and Ehrenfest's Theorem

The following unitary operators generate the continuous transformations relevant to the system.

2.4.1 Translation and Evolution Operators

- **Spatial Translation:** Generated by the momentum operator $\hat{\mathbf{p}}$. For a finite vector $\mathbf{a}$ and an infinitesimal one $d\mathbf{a}$:

$$\hat{T}(\mathbf{a}) = \exp\left(-\frac{i}{\hbar} \mathbf{a} \cdot \hat{\mathbf{p}}\right), \quad \hat{T}(d\mathbf{a}) \approx \hat{\mathbb{I}} - \frac{i}{\hbar} d\mathbf{a} \cdot \hat{\mathbf{p}}. \quad (11)$$

- **Time Evolution:** Generated by the Hamiltonian $\hat{H}$.

$$\hat{U}(t) = \exp\left(-\frac{i}{\hbar}\hat{H}t\right), \quad \hat{U}(dt) \approx \hat{\mathbb{I}} - \frac{i}{\hbar}\hat{H}dt. \quad (12)$$

- **Momentum Space Translation:** Analogous to spatial translation, the position operator $\hat{\mathbf{r}}$ generates displacements in momentum space (boosts). For a momentum displacement $\mathbf{q}$:

$$\hat{T}_{\mathbf{p}}(\mathbf{q}) = \exp\left(\frac{i}{\hbar}\mathbf{q} \cdot \hat{\mathbf{r}}\right). \quad (13)$$

The action of these operators defines the change of basis between the Hilbert spaces associated with the observables.

2.4.2 Correspondence Principle and Ehrenfest's Theorem

The connection to classical mechanics is formally established via Ehrenfest's Theorem, which governs the time evolution of expectation values. For an operator $\hat{A}$ with no explicit time dependence:

$$\frac{d}{dt}\langle\hat{A}\rangle = \frac{1}{i\hbar}\langle[\hat{A}, \hat{H}]\rangle. \quad (14)$$

Applied to the free particle ($\hat{H} = \hat{p}^2/2m$):

- **Position:** $\frac{d}{dt}\langle\hat{\mathbf{r}}\rangle = \frac{1}{m}\langle\hat{\mathbf{p}}\rangle$ (Velocity-Momentum relation).
- **Momentum:** $\frac{d}{dt}\langle\hat{\mathbf{p}}\rangle = 0$ (Conservation of linear momentum due to translational invariance).

These results ensure that wave packets follow classical trajectories in the macroscopic limit, satisfying the correspondence principle.

3 Definition of the Angular Momentum Operator

In classical mechanics, the angular momentum of a particle with position vector $\mathbf{r}$ and linear momentum $\mathbf{p}$ is defined as the vector product $\mathbf{L}_{cl} = \mathbf{r} \times \mathbf{p}$. To transition to quantum mechanics, we apply the correspondence principle, promoting the dynamical variables to linear operators acting on the Hilbert space: $\mathbf{r} \rightarrow \hat{\mathbf{r}}$ and $\mathbf{p} \rightarrow \hat{\mathbf{p}}$.

However, since $\hat{\mathbf{r}}$ and $\hat{\mathbf{p}}$ are non-commuting operators, the definition of the vector product must be handled with algebraic rigor to ensure the resulting operator is Hermitian (measurable) and unambiguous.

3.1 The Cross Product of Vector Operators

Let $\hat{\mathbf{A}}$ and $\hat{\mathbf{B}}$ be two vector operators in $\mathbb{R}^3$. The i -th component of their cross product, $(\hat{\mathbf{A}} \times \hat{\mathbf{B}})_i$, is defined using the Levi-Civita completely antisymmetric tensor ϵ_{ijk} , where indices i, j, k run over $\{1, 2, 3\}$ (identifying $1 \rightarrow x, 2 \rightarrow y, 3 \rightarrow z$):

$$(\hat{\mathbf{A}} \times \hat{\mathbf{B}})_i = \sum_{j,k} \epsilon_{ijk} \hat{A}_j \hat{B}_k. \quad (15)$$

Applying this definition to the position and momentum operators, we define the orbital angular momentum operator $\hat{\mathbf{L}}$ as:

$$\hat{\mathbf{L}} \equiv \hat{\mathbf{r}} \times \hat{\mathbf{p}}. \quad (16)$$

In component form, this definition implies:

$$\hat{L}_i = \epsilon_{ijk} \hat{x}_j \hat{p}_k, \quad (17)$$

where we have adopted the Einstein summation convention for repeated indices j and k .

3.2 Non-Commutativity and Hermiticity

A fundamental postulate of quantum mechanics is that physical observables are represented by Hermitian operators ($\hat{\mathcal{O}} = \hat{\mathcal{O}}^\dagger$). In general, the product of two Hermitian operators $\hat{A}\hat{B}$ is not Hermitian unless $[\hat{A}, \hat{B}] = 0$. Since position and momentum do not commute ($[\hat{x}_j, \hat{p}_k] = i\hbar\delta_{jk}$), we must verify that the definition in Eq. (17) yields a Hermitian operator without the need for manual symmetrization.

We take the adjoint of the i -th component:

$$\hat{L}_i^\dagger = (\epsilon_{ijk} \hat{x}_j \hat{p}_k)^\dagger. \quad (18)$$

Using the property $(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger \hat{A}^\dagger$ and the fact that $\hat{\mathbf{x}}$ and $\hat{\mathbf{p}}$ are Hermitian ($\hat{x}_j^\dagger = \hat{x}_j$, $\hat{p}_k^\dagger = \hat{p}_k$) and ϵ_{ijk} is real:

$$\hat{L}_i^\dagger = \epsilon_{ijk} \hat{p}_k^\dagger \hat{x}_j^\dagger = \epsilon_{ijk} \hat{p}_k \hat{x}_j. \quad (19)$$

Notice the order of operators has reversed. To restore the original order, we use the canonical commutation relation:

$$[\hat{x}_j, \hat{p}_k] = i\hbar\delta_{jk} \implies \hat{p}_k \hat{x}_j = \hat{x}_j \hat{p}_k - i\hbar\delta_{jk}. \quad (20)$$

Substituting this back into the expression for $\hat{L}_i^\dagger$:

$$\hat{L}_i^\dagger = \epsilon_{ijk} (\hat{x}_j \hat{p}_k - i\hbar\delta_{jk}) \quad (21)$$

$$= \epsilon_{ijk} \hat{x}_j \hat{p}_k - i\hbar\epsilon_{ijk} \delta_{jk}. \quad (22)$$

The second term contains the contraction of the antisymmetric tensor ϵ_{ijk} with the symmetric Kronecker delta δ_{jk} . Since ϵ_{ijk} vanishes if any two indices are equal (which δ_{jk} forces), we have:

$$\epsilon_{ijk} \delta_{jk} = \epsilon_{ijj} = 0. \quad (23)$$

Therefore:

$$\hat{L}_i^\dagger = \epsilon_{ijk} \hat{x}_j \hat{p}_k = \hat{L}_i. \quad (24)$$

Conclusion: The operator $\hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}}$ is inherently Hermitian. The specific non-commutativity of $\hat{\mathbf{r}}$ and $\hat{\mathbf{p}}$ does not affect the hermiticity of $\hat{\mathbf{L}}$ because the cross product couples orthogonal components (e.g., x and p_y) which mutually commute ($[\hat{x}, \hat{p}_y] = 0$).

3.3 Explicit Component Form

Expanding the tensor notation ($\hat{L}_x$ corresponds to $i = 1$, etc.), we obtain the explicit forms used in standard calculations:

$$\hat{L}_x = \hat{y}\hat{p}_z - \hat{z}\hat{p}_y, \quad (25)$$

$$\hat{L}_y = \hat{z}\hat{p}_x - \hat{x}\hat{p}_z, \quad (26)$$

$$\hat{L}_z = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x. \quad (27)$$

These expressions confirm that the quantum angular momentum operator preserves the functional form of its classical counterpart, provided the order of operators is maintained as $\mathbf{r} \times \mathbf{p}$.

3.4 Geometric Algebra Formulation: Derivation from the Fundamental Product

A more profound insight into the nature of angular momentum can be obtained using Geometric Algebra (Clifford Algebra $\mathcal{C}\ell_{3,0}$). In this framework, the fundamental operation is the **Geometric Product**, denoted by the juxtaposition of vectors $\hat{\mathbf{r}}\hat{\mathbf{p}}$.

For any two vectors, the geometric product decomposes into a symmetric scalar part (Grade 0) and an antisymmetric bivector part (Grade 2):

$$\hat{\mathbf{r}}\hat{\mathbf{p}} = \hat{\mathbf{r}} \cdot \hat{\mathbf{p}} + \hat{\mathbf{r}} \wedge \hat{\mathbf{p}}. \quad (28)$$

Here, $\hat{\mathbf{r}} \cdot \hat{\mathbf{p}}$ is the standard scalar product (related to dilation), and $\hat{\mathbf{r}} \wedge \hat{\mathbf{p}}$ is the **exterior product** (wedge product).

3.4.1 Isolating the Bivector of Rotation

The angular momentum is physically defined as the generator of rotations. In Geometric Algebra, rotations are generated by bivectors. To isolate this physical quantity from the fundamental geometric product, we employ the **grade projection operator** $\langle \dots \rangle_k$:

$$\hat{\mathcal{L}} \equiv \langle \hat{\mathbf{r}}\hat{\mathbf{p}} \rangle_2 = \hat{\mathbf{r}} \wedge \hat{\mathbf{p}}. \quad (29)$$

Unlike the vector cross product, which is limited to three dimensions, the bivector $\hat{\mathcal{L}}$ represents the *oriented plane* of rotation and is valid in any dimension.

3.4.2 Relation to the Cross Product (Hodge Dual)

The assignment requires the definition $\hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}}$. This vector $\hat{\mathbf{L}}$ is merely the **Hodge dual** of the fundamental bivector $\hat{\mathcal{L}}$. In 3D Euclidean space, the unit pseudoscalar $I = \hat{e}_1\hat{e}_2\hat{e}_3$ maps bivectors to vectors via the duality relation:

$$\hat{\mathbf{L}} = -iI(\hat{\mathbf{r}} \wedge \hat{\mathbf{p}}). \quad (30)$$

Thus, the standard operator $\hat{\mathbf{L}}$ is the vector normal to the bivector plane $\hat{\mathcal{L}}$. While both definitions are mathematically consistent in 3D, the geometric product formulation $\hat{\mathbf{r}}\hat{\mathbf{p}}$ reveals that angular momentum arises naturally as the antisymmetric component of the interaction between position and momentum.

4 Representation of $\hat{\mathbf{L}}$ in Position Space

In the position representation, the Hilbert space is spanned by the eigenkets $|\mathbf{r}\rangle$. The position operator acts multiplicatively, $\hat{\mathbf{r}} \rightarrow \mathbf{r}$, and the linear momentum operator acts as the gradient generator.

4.1 Representation in Cartesian Coordinates

4.1.1 Derivation from Fundamental Operators

Using the position representation of the momentum operator[cite: 42]:

$$\hat{\mathbf{p}} = -i\hbar\nabla_{\mathbf{r}}, \quad (31)$$

we substitute this into the definition of angular momentum derived in the previous section ($\hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}}$):

$$\hat{\mathbf{L}} = \hat{\mathbf{r}} \times (-i\hbar\nabla) = -i\hbar(\mathbf{r} \times \nabla). \quad (32)$$

4.1.2 Explicit Components

Expanding the cross product in Cartesian coordinates using the determinant mnemonic (or the Levi-Civita definition):

$$\mathbf{r} \times \nabla = \begin{vmatrix} \hat{\mathbf{e}}_x & \hat{\mathbf{e}}_y & \hat{\mathbf{e}}_z \\ x & y & z \\ \partial_x & \partial_y & \partial_z \end{vmatrix} = \hat{\mathbf{e}}_x(y\partial_z - z\partial_y) - \hat{\mathbf{e}}_y(x\partial_z - z\partial_x) + \hat{\mathbf{e}}_z(x\partial_y - y\partial_x). \quad (33)$$

Multiplying by the factor $-i\hbar$, we obtain the explicit components[cite: 45]:

$$\hat{L}_x = -i\hbar \left(y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y} \right), \quad (34)$$

$$\hat{L}_y = -i\hbar \left(z \frac{\partial}{\partial x} - x \frac{\partial}{\partial z} \right), \quad (35)$$

$$\hat{L}_z = -i\hbar \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right). \quad (36)$$

4.2 Representation in Spherical Coordinates

We now transform these operators to spherical coordinates (r, θ, ϕ) defined by:

$$x = r \sin \theta \cos \phi,$$

$$y = r \sin \theta \sin \phi,$$

$$z = r \cos \theta.$$

4.2.1 Derivation of $\hat{L}_z$

To find the expression for $\hat{L}_z$ in spherical coordinates, we apply the Chain Rule to the partial derivative with respect to ϕ . Let $\psi(x, y, z)$ be a function of position. The derivative with respect to ϕ is:

$$\frac{\partial}{\partial \phi} = \frac{\partial x}{\partial \phi} \frac{\partial}{\partial x} + \frac{\partial y}{\partial \phi} \frac{\partial}{\partial y} + \frac{\partial z}{\partial \phi} \frac{\partial}{\partial z}. \quad (37)$$

Calculating the partial derivatives of the coordinates:

- $\frac{\partial x}{\partial \phi} = \frac{\partial}{\partial \phi}(r \sin \theta \cos \phi) = -r \sin \theta \sin \phi = -y.$
- $\frac{\partial y}{\partial \phi} = \frac{\partial}{\partial \phi}(r \sin \theta \sin \phi) = r \sin \theta \cos \phi = x.$
- $\frac{\partial z}{\partial \phi} = 0$ (since z does not depend on ϕ).

Substituting these back into the chain rule expression:

$$\frac{\partial}{\partial \phi} = (-y) \frac{\partial}{\partial x} + (x) \frac{\partial}{\partial y} = x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x}. \quad (38)$$

Comparing this result with Eq. (36), we immediately identify the term in parentheses. Thus:

$$\hat{L}_z = -i\hbar \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right) = -i\hbar \frac{\partial}{\partial \phi}. \quad (39)$$

4.2.2 Derivation of $\hat{L}^2$

To derive the squared total angular momentum $\hat{L}^2 = \hat{\mathbf{L}} \cdot \hat{\mathbf{L}}$, we avoid the tedious squaring of Cartesian components and instead use the vector properties of the Gradient in spherical coordinates.

Step 1: The Gradient in Spherical Coordinates The gradient operator ∇ is expressed in the spherical orthonormal basis ($\hat{\mathbf{e}}_r, \hat{\mathbf{e}}_\theta, \hat{\mathbf{e}}_\phi$) as:

$$\nabla = \hat{\mathbf{e}}_r \frac{\partial}{\partial r} + \hat{\mathbf{e}}_\theta \frac{1}{r} \frac{\partial}{\partial \theta} + \hat{\mathbf{e}}_\phi \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi}. \quad (40)$$

Step 2: The Vector $\hat{\mathbf{L}}$ in Spherical Basis Using $\mathbf{r} = r\hat{\mathbf{e}}_r$ and calculating the cross product $\hat{\mathbf{L}} = -i\hbar(\mathbf{r} \times \nabla)$:

$$\hat{\mathbf{L}} = -i\hbar \left[r\hat{\mathbf{e}}_r \times \left(\hat{\mathbf{e}}_r \frac{\partial}{\partial r} + \hat{\mathbf{e}}_\theta \frac{1}{r} \frac{\partial}{\partial \theta} + \hat{\mathbf{e}}_\phi \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} \right) \right]. \quad (41)$$

Using the orthogonality relations $\hat{\mathbf{e}}_r \times \hat{\mathbf{e}}_r = 0$, $\hat{\mathbf{e}}_r \times \hat{\mathbf{e}}_\theta = \hat{\mathbf{e}}_\phi$, and $\hat{\mathbf{e}}_r \times \hat{\mathbf{e}}_\phi = -\hat{\mathbf{e}}_\theta$:

$$\begin{aligned}\hat{\mathbf{L}} &= -i\hbar \left[\left(r \frac{1}{r} \right) (\hat{\mathbf{e}}_r \times \hat{\mathbf{e}}_\theta) \frac{\partial}{\partial \theta} + \left(r \frac{1}{r \sin \theta} \right) (\hat{\mathbf{e}}_r \times \hat{\mathbf{e}}_\phi) \frac{\partial}{\partial \phi} \right] \\ &= -i\hbar \left(\hat{\mathbf{e}}_\phi \frac{\partial}{\partial \theta} - \hat{\mathbf{e}}_\theta \frac{1}{\sin \theta} \frac{\partial}{\partial \phi} \right).\end{aligned}\quad (42)$$

Step 3: Calculating $\hat{L}^2 = \hat{\mathbf{L}} \cdot \hat{\mathbf{L}}$ We apply the vector operator $\hat{\mathbf{L}}$ from Eq. (42) to itself. Extreme care must be taken because the unit vectors $\hat{\mathbf{e}}_\theta$ and $\hat{\mathbf{e}}_\phi$ depend on the coordinates, so derivatives acting on them are non-zero.

$$\hat{L}^2 = (-i\hbar)^2 \left(\hat{\mathbf{e}}_\phi \frac{\partial}{\partial \theta} - \frac{\hat{\mathbf{e}}_\theta}{\sin \theta} \frac{\partial}{\partial \phi} \right) \cdot \left(\hat{\mathbf{e}}_\phi \frac{\partial}{\partial \theta} - \frac{\hat{\mathbf{e}}_\theta}{\sin \theta} \frac{\partial}{\partial \phi} \right). \quad (43)$$

Expanding the dot product (remembering that the first operator acts on everything to its right, including unit vectors):

$$\begin{aligned}\frac{\hat{L}^2}{-\hbar^2} &= \underbrace{\hat{\mathbf{e}}_\phi \cdot \frac{\partial}{\partial \theta} \left(\hat{\mathbf{e}}_\phi \frac{\partial}{\partial \theta} \right)}_{\text{Term 1}} - \underbrace{\hat{\mathbf{e}}_\phi \cdot \frac{\partial}{\partial \theta} \left(\frac{\hat{\mathbf{e}}_\theta}{\sin \theta} \frac{\partial}{\partial \phi} \right)}_{\text{Term 2}} - \underbrace{\frac{\hat{\mathbf{e}}_\theta}{\sin \theta} \cdot \frac{\partial}{\partial \phi} \left(\hat{\mathbf{e}}_\phi \frac{\partial}{\partial \theta} \right)}_{\text{Term 3}} + \underbrace{\frac{\hat{\mathbf{e}}_\theta}{\sin \theta} \cdot \frac{\partial}{\partial \phi} \left(\frac{\hat{\mathbf{e}}_\theta}{\sin \theta} \frac{\partial}{\partial \phi} \right)}_{\text{Term 4}}.\end{aligned}\quad (44)$$

Required Derivatives of Unit Vectors:

- $\partial_\theta \hat{\mathbf{e}}_\phi = 0$.
- $\partial_\theta \hat{\mathbf{e}}_\theta = -\hat{\mathbf{e}}_r$ (Orthogonal to $\hat{\mathbf{e}}_\phi$, so dot product vanishes).
- $\partial_\phi \hat{\mathbf{e}}_\phi = -\sin \theta \hat{\mathbf{e}}_r - \cos \theta \hat{\mathbf{e}}_\theta$.
- $\partial_\phi \hat{\mathbf{e}}_\theta = \cos \theta \hat{\mathbf{e}}_\phi$.

Evaluating Terms:

Term 1: Since $\partial_\theta \hat{\mathbf{e}}_\phi = 0$:

$$\hat{\mathbf{e}}_\phi \cdot \left(\hat{\mathbf{e}}_\phi \frac{\partial^2}{\partial \theta^2} \right) = \frac{\partial^2}{\partial \theta^2}.$$

Term 2: Contains $\hat{\mathbf{e}}_\phi \cdot \partial_\theta \hat{\mathbf{e}}_\theta$. Since $\partial_\theta \hat{\mathbf{e}}_\theta = -\hat{\mathbf{e}}_r$, the dot product with $\hat{\mathbf{e}}_\phi$ is 0. Cross terms vanish here.

Term 3:

$$\frac{\hat{\mathbf{e}}_\theta}{\sin \theta} \cdot \left((\partial_\phi \hat{\mathbf{e}}_\phi) \frac{\partial}{\partial \theta} + \hat{\mathbf{e}}_\phi \frac{\partial^2}{\partial \phi \partial \theta} \right).$$

Using $\partial_\phi \hat{\mathbf{e}}_\phi = -\sin \theta \hat{\mathbf{e}}_r - \cos \theta \hat{\mathbf{e}}_\theta$:

$$\frac{\hat{\mathbf{e}}_\theta}{\sin \theta} \cdot \left(-\cos \theta \hat{\mathbf{e}}_\theta \frac{\partial}{\partial \theta} \right) = -\frac{\cos \theta}{\sin \theta} \frac{\partial}{\partial \theta} = -\cot \theta \frac{\partial}{\partial \theta}.$$

Term 4:

$$\frac{\hat{\mathbf{e}}_\theta}{\sin \theta} \cdot \frac{\partial}{\partial \phi} \left(\frac{\hat{\mathbf{e}}_\theta}{\sin \theta} \frac{\partial}{\partial \phi} \right).$$

The derivative acts on $\hat{\mathbf{e}}_\theta$ and the function. However, $\partial_\phi \hat{\mathbf{e}}_\theta = \cos \theta \hat{\mathbf{e}}_\phi$, which is orthogonal to the pre-factor $\hat{\mathbf{e}}_\theta$. Thus, only the derivative on the scalar function survives:

$$\frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2}.$$

Assembly: Summing the non-vanishing contributions:

$$\frac{\hat{L}^2}{-\hbar^2} = \frac{\partial^2}{\partial \theta^2} + \cot \theta \frac{\partial}{\partial \theta} + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2}. \quad (45)$$

Recognizing that $\frac{\partial^2}{\partial \theta^2} + \cot \theta \frac{\partial}{\partial \theta} = \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right)$, we arrive at the final expression[cite: 48]:

$$\hat{L}^2 = -\hbar^2 \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right]. \quad (46)$$

5 Angular Momentum as the Generator of Rotations

In this section, we establish the fundamental role of the angular momentum operator $\hat{\mathbf{L}}$ as the generator of spatial rotations. We proceed from the definition of the unitary rotation operator to the conservation laws derived from the symmetry of the Hamiltonian.

5.1 Finite Rotation Operator

5.1.1 Definition

We define the rotation of a physical system by an angle $|\boldsymbol{\theta}|$ around an axis defined by the unit vector $\mathbf{n} = \boldsymbol{\theta}/|\boldsymbol{\theta}|$ using the exponential map. The unitary operator for a finite rotation is defined as[cite: 53]:

$$\hat{R}(\boldsymbol{\theta}) = \exp \left(-\frac{i}{\hbar} \boldsymbol{\theta} \cdot \hat{\mathbf{L}} \right). \quad (47)$$

Since $\hat{\mathbf{L}}$ is Hermitian, $\hat{R}(\boldsymbol{\theta})$ is unitary ($\hat{R}^\dagger = \hat{R}^{-1}$), ensuring the conservation of probability (norm of the state) during rotation.

5.1.2 Action on Position Kets

To demonstrate the action of $\hat{R}(\boldsymbol{\theta})$ on the position eigenkets $|\mathbf{r}\rangle$, we consider the transformation of the position operator $\hat{\mathbf{r}}$. Using the Baker-Campbell-Hausdorff lemma or the infinitesimal properties derived below, one can show that:

$$\hat{R}^\dagger(\boldsymbol{\theta}) \hat{\mathbf{r}} \hat{R}(\boldsymbol{\theta}) = \mathcal{R}(\boldsymbol{\theta}) \hat{\mathbf{r}}, \quad (48)$$

where $\mathcal{R}(\boldsymbol{\theta})$ is the classical 3×3 rotation matrix. Applying this to a position eigenket $|\mathbf{r}\rangle$:

$$\hat{\mathbf{r}} \left(\hat{R}(\boldsymbol{\theta}) |\mathbf{r}\rangle \right) = \hat{R}(\boldsymbol{\theta}) \left(\hat{R}^\dagger \hat{\mathbf{r}} \hat{R} \right) |\mathbf{r}\rangle = \hat{R}(\boldsymbol{\theta}) (\mathcal{R}\mathbf{r}) |\mathbf{r}\rangle. \quad (49)$$

Since $|\mathbf{r}\rangle$ is an eigenstate of $\hat{\mathbf{r}}$ with eigenvalue $\mathbf{r}$, the rotated ket must be an eigenstate of the position operator with eigenvalue $\mathcal{R}\mathbf{r}$. Thus[cite: 55]:

$$\hat{R}(\boldsymbol{\theta}) |\mathbf{r}\rangle = |\mathcal{R}(\boldsymbol{\theta})\mathbf{r}\rangle. \quad (50)$$

Physically, this implies that the operator actively transports a particle localized at $\mathbf{r}$ to the rotated point $\mathcal{R}\mathbf{r}$.

5.2 Infinitesimal Rotation

5.2.1 First-Order Expansion

Consider an infinitesimal rotation by an angle $d\boldsymbol{\theta}$. Expanding the exponential to first order in the parameter $d\boldsymbol{\theta}$ [cite: 59]:

$$\hat{R}(d\boldsymbol{\theta}) \approx \hat{\mathbb{I}} - \frac{i}{\hbar} d\boldsymbol{\theta} \cdot \hat{\mathbf{L}}. \quad (51)$$

5.2.2 Action on the Wavefunction

We examine how this operator transforms the wavefunction $\psi(\mathbf{r}) = \langle \mathbf{r} | \psi \rangle$. The transformed state $|\psi'\rangle = \hat{R}(d\boldsymbol{\theta}) |\psi\rangle$ has the wavefunction:

$$\psi'(\mathbf{r}) = \langle \mathbf{r} | \hat{R}(d\boldsymbol{\theta}) |\psi\rangle \approx \langle \mathbf{r} | \left(1 - \frac{i}{\hbar} d\boldsymbol{\theta} \cdot \hat{\mathbf{L}} \right) |\psi\rangle. \quad (52)$$

Distributing the bra-ket:

$$(\hat{R}(d\boldsymbol{\theta})\psi)(\mathbf{r}) \approx \psi(\mathbf{r}) - \frac{i}{\hbar} d\boldsymbol{\theta} \cdot \langle \mathbf{r} | \hat{\mathbf{L}} | \psi \rangle. \quad (53)$$

Using the position representation of angular momentum derived in Section 3, $(\hat{\mathbf{L}}\psi)(\mathbf{r}) = -i\hbar(\mathbf{r} \times \nabla)\psi(\mathbf{r})$, we substitute:

$$(\hat{R}(d\boldsymbol{\theta})\psi)(\mathbf{r}) \approx \psi(\mathbf{r}) - \frac{i}{\hbar} d\boldsymbol{\theta} \cdot [-i\hbar(\mathbf{r} \times \nabla)\psi(\mathbf{r})]. \quad (54)$$

$$(\hat{R}(d\boldsymbol{\theta})\psi)(\mathbf{r}) \approx \psi(\mathbf{r}) - d\boldsymbol{\theta} \cdot (\mathbf{r} \times \nabla)\psi(\mathbf{r}). \quad (55)$$

Using the vector identity $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \times \mathbf{B}) \cdot \mathbf{C}$, we rewrite the correction term:

$$d\boldsymbol{\theta} \cdot (\mathbf{r} \times \nabla) = (d\boldsymbol{\theta} \times \mathbf{r}) \cdot \nabla. \quad (56)$$

Recognizing that $d\mathbf{r} = d\boldsymbol{\theta} \times \mathbf{r}$ is the displacement vector due to rotation, we obtain the Taylor expansion for a scalar field under infinitesimal coordinate displacement[cite: 61]:

$$\psi'(\mathbf{r}) \approx \psi(\mathbf{r}) - d\mathbf{r} \cdot \nabla \psi(\mathbf{r}) \approx \psi(\mathbf{r} - d\mathbf{r}). \quad (57)$$

This confirms consistency with the passive transformation view: rotating the state active forward is equivalent to evaluating the original function at the reverse-rotated coordinates.

5.3 Symmetries and Conservation Laws

5.3.1 Commutation with the Free Hamiltonian

The free particle Hamiltonian is $\hat{H} = \frac{\hat{p}^2}{2m}$. To test for rotational invariance, we compute the commutator $[\hat{H}, \hat{L}_i]$ [cite: 64]. Since $\hat{p}^2$ is a scalar operator (dot product of a vector operator with itself), it should commute with the generator of rotations. Explicitly:

$$[\hat{L}_i, \hat{p}^2] = \sum_j [\hat{L}_i, \hat{p}_j \hat{p}_j] = \sum_j \left(\hat{p}_j [\hat{L}_i, \hat{p}_j] + [\hat{L}_i, \hat{p}_j] \hat{p}_j \right). \quad (58)$$

Using the canonical commutation relation for angular momentum and vector operators, $[\hat{L}_i, \hat{p}_j] = i\hbar \epsilon_{ijk} \hat{p}_k$:

$$[\hat{L}_i, \hat{p}^2] = i\hbar \sum_{j,k} \epsilon_{ijk} (\hat{p}_j \hat{p}_k + \hat{p}_k \hat{p}_j) = 2i\hbar \sum_{j,k} \epsilon_{ijk} \hat{p}_j \hat{p}_k. \quad (59)$$

This sum involves the contraction of the antisymmetric tensor ϵ_{ijk} (in indices j, k) with the symmetric tensor $\hat{p}_j \hat{p}_k$. Therefore, the sum vanishes identically:

$$[\hat{H}, \hat{\mathbf{L}}] = 0. \quad (60)$$

5.3.2 Conservation of Angular Momentum

We invoke Ehrenfest's Theorem, established in Section 1[cite: 32]:

$$\frac{d}{dt} \langle \hat{\mathbf{L}} \rangle = \frac{1}{i\hbar} \langle [\hat{\mathbf{L}}, \hat{H}] \rangle. \quad (61)$$

Substituting the zero commutator derived above:

$$\frac{d}{dt} \langle \hat{\mathbf{L}} \rangle = 0. \quad (62)$$

This result [cite: 66] constitutes the quantum mechanical proof of the conservation of angular momentum for a free particle.

5.3.3 Connection to the Correspondence Principle

Classically, for a free particle, there is no torque ($\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F} = 0$), implying $d\mathbf{L}_{cl}/dt = 0$. The quantum result perfectly mirrors the classical law, as guaranteed by the Correspondence Principle. The rotational symmetry of space (isotropy) leads to the conservation of angular momentum in both frameworks, with the operator $\hat{\mathbf{L}}$ playing the dual role of the observable quantity and the generator of the symmetry group $SO(3)$.

6 Angular Momentum Commutation Algebra

6.1 Fundamental Commutators

To establish the Lie algebra of the rotation group $SO(3)$, we must compute the commutation relations between the components of the angular momentum operator explicitly. We start from the canonical commutation relations for position and momentum:

$$[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}, \quad [\hat{x}_i, \hat{x}_j] = 0, \quad [\hat{p}_i, \hat{p}_j] = 0. \quad (63)$$

Using the explicit definitions derived in the previous section:

$$\hat{L}_x = \hat{y}\hat{p}_z - \hat{z}\hat{p}_y, \quad \hat{L}_y = \hat{z}\hat{p}_x - \hat{x}\hat{p}_z, \quad \hat{L}_z = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x. \quad (64)$$

6.1.1 Commutator $[\hat{L}_x, \hat{L}_y]$

We substitute the explicit expressions for the operators:

$$[\hat{L}_x, \hat{L}_y] = [(\hat{y}\hat{p}_z - \hat{z}\hat{p}_y), (\hat{z}\hat{p}_x - \hat{x}\hat{p}_z)] \quad (65)$$

Using the linearity of the commutator $[A + B, C + D] = [A, C] + [A, D] + [B, C] + [B, D]$, we expand this expression into four terms:

$$[\hat{L}_x, \hat{L}_y] = \underbrace{[\hat{y}\hat{p}_z, \hat{z}\hat{p}_x]}_{\text{I}} - \underbrace{[\hat{y}\hat{p}_z, \hat{x}\hat{p}_z]}_{\text{II}} - \underbrace{[\hat{z}\hat{p}_y, \hat{z}\hat{p}_x]}_{\text{III}} + \underbrace{[\hat{z}\hat{p}_y, \hat{x}\hat{p}_z]}_{\text{IV}} \quad (66)$$

We analyze each term separately using the product rule $[AB, CD] = A[B, C]D + AC[B, D] + [A, C]BD + C[A, D]B$ and the fact that operators acting on different Cartesian coordinates commute (e.g., $[\hat{y}, \hat{p}_x] = 0$).

- **Term I:** $[\hat{y}\hat{p}_z, \hat{z}\hat{p}_x]$. Here, $\hat{y}$ and $\hat{p}_x$ commute with everything in the bracket. The non-commuting pair is $\hat{p}_z$ and $\hat{z}$.

$$[\hat{y}\hat{p}_z, \hat{z}\hat{p}_x] = \hat{y}\hat{p}_x[\hat{p}_z, \hat{z}] = \hat{y}\hat{p}_x(-i\hbar) = -i\hbar\hat{y}\hat{p}_x \quad (67)$$

(Recall that $[\hat{z}, \hat{p}_z] = i\hbar$, so $[\hat{p}_z, \hat{z}] = -i\hbar$).

- **Term II:** $[\hat{y}\hat{p}_z, \hat{x}\hat{p}_z]$. All operators involve different coordinates (y, z, x) or the same operator ($\hat{p}_z$ with $\hat{p}_z$), so they all commute.

$$[\hat{y}\hat{p}_z, \hat{x}\hat{p}_z] = 0 \quad (68)$$

- **Term III:** $[\hat{z}\hat{p}_y, \hat{z}\hat{p}_x]$. Similarly, $\hat{z}$ commutes with $\hat{z}$, and $\hat{p}_y$ commutes with $\hat{p}_x$.

$$[\hat{z}\hat{p}_y, \hat{z}\hat{p}_x] = 0 \quad (69)$$

- **Term IV:** $[\hat{z}\hat{p}_y, \hat{x}\hat{p}_z]$. Here, $\hat{p}_y$ and $\hat{x}$ commute with the others. The non-commuting pair is $\hat{z}$ and $\hat{p}_z$.

$$[\hat{z}\hat{p}_y, \hat{x}\hat{p}_z] = \hat{x}\hat{p}_y[\hat{z}, \hat{p}_z] = \hat{x}\hat{p}_y(i\hbar) = i\hbar\hat{x}\hat{p}_y \quad (70)$$

Substituting these results back into the expansion:

$$[\hat{L}_x, \hat{L}_y] = (-i\hbar\hat{y}\hat{p}_x) - 0 - 0 + (i\hbar\hat{x}\hat{p}_y) \quad (71)$$

$$[\hat{L}_x, \hat{L}_y] = i\hbar(\hat{x}\hat{p}_y - \hat{y}\hat{p}_x) \quad (72)$$

Recognizing the term in parentheses as $\hat{L}_z$, we arrive at the first fundamental relation:

$$[\hat{L}_x, \hat{L}_y] = i\hbar\hat{L}_z \quad (73)$$

6.1.2 Commutator $[\hat{L}_y, \hat{L}_z]$

Substituting the explicit forms:

$$[\hat{L}_y, \hat{L}_z] = [(\hat{z}\hat{p}_x - \hat{x}\hat{p}_z), (\hat{x}\hat{p}_y - \hat{y}\hat{p}_x)] \quad (74)$$

Expanding the terms similarly:

$$[\hat{z}\hat{p}_x, \hat{x}\hat{p}_y] = \hat{z}\hat{p}_y[\hat{p}_x, \hat{x}] = \hat{z}\hat{p}_y(-i\hbar) \quad (75)$$

$$[\hat{z}\hat{p}_x, \hat{y}\hat{p}_x] = 0 \quad (76)$$

$$[\hat{x}\hat{p}_z, \hat{x}\hat{p}_y] = 0 \quad (77)$$

$$[\hat{x}\hat{p}_z, \hat{y}\hat{p}_x] = \hat{y}\hat{p}_z[\hat{x}, \hat{p}_x] = \hat{y}\hat{p}_z(i\hbar) \quad (78)$$

Reassembling (paying attention to the negative signs in the original expansion):

$$[\hat{L}_y, \hat{L}_z] = (-i\hbar\hat{z}\hat{p}_y) - (0) - (0) + (-1)(-1)(i\hbar\hat{y}\hat{p}_z) \quad (79)$$

$$[\hat{L}_y, \hat{L}_z] = i\hbar(\hat{y}\hat{p}_z - \hat{z}\hat{p}_y) = i\hbar\hat{L}_x \quad (80)$$

6.1.3 Commutator $[\hat{L}_z, \hat{L}_x]$

Substituting the explicit forms:

$$[\hat{L}_z, \hat{L}_x] = [(\hat{x}\hat{p}_y - \hat{y}\hat{p}_x), (\hat{y}\hat{p}_z - \hat{z}\hat{p}_y)] \quad (81)$$

Expanding the terms:

$$[\hat{x}\hat{p}_y, \hat{y}\hat{p}_z] = \hat{x}\hat{p}_z[\hat{p}_y, \hat{y}] = \hat{x}\hat{p}_z(-i\hbar) \quad (82)$$

$$[\hat{x}\hat{p}_y, \hat{z}\hat{p}_y] = 0 \quad (83)$$

$$[\hat{y}\hat{p}_x, \hat{y}\hat{p}_z] = 0 \quad (84)$$

$$[\hat{y}\hat{p}_x, \hat{z}\hat{p}_y] = \hat{z}\hat{p}_x[\hat{y}, \hat{p}_y] = \hat{z}\hat{p}_x(i\hbar) \quad (85)$$

Reassembling:

$$[\hat{L}_z, \hat{L}_x] = (-i\hbar\hat{x}\hat{p}_z) - 0 - 0 - (-1)(i\hbar\hat{z}\hat{p}_x) \quad (86)$$

$$[\hat{L}_z, \hat{L}_x] = i\hbar(\hat{z}\hat{p}_x - \hat{x}\hat{p}_z) = i\hbar\hat{L}_y \quad (87)$$

These three results can be summarized compactly using the Levi-Civita antisymmetric tensor, confirming that the angular momentum components satisfy the Lie algebra of $SO(3)$:

$$[\hat{L}_i, \hat{L}_j] = i\hbar\epsilon_{ijk}\hat{L}_k \quad (88)$$

6.2 The Operator $\hat{L}^2$ and its Commutators

6.2.1 Definition

We define the operator corresponding to the square of the magnitude of the angular momentum vector as the sum of the squares of its Cartesian components:

$$\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2 \quad (89)$$

6.2.2 Commutation with Components

We now wish to demonstrate that the scalar operator $\hat{L}^2$ commutes with any specific component of the vector operator $\hat{L}$. Without loss of generality, we choose to calculate the commutator with $\hat{L}_z$. Due to the rotational symmetry of the indices, the result will be valid for $\hat{L}_x$ and $\hat{L}_y$ as well.

$$[\hat{L}^2, \hat{L}_z] = [\hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2, \hat{L}_z] \quad (90)$$

Using the linearity of the commutator, we separate this into three terms:

$$[\hat{L}^2, \hat{L}_z] = [\hat{L}_x^2, \hat{L}_z] + [\hat{L}_y^2, \hat{L}_z] + [\hat{L}_z^2, \hat{L}_z] \quad (91)$$

The last term vanishes trivially because any operator commutes with itself and powers of itself ($[\hat{A}^n, \hat{A}] = 0$):

$$[\hat{L}_z^2, \hat{L}_z] = 0 \quad (92)$$

For the remaining terms, we use the property $[A^2, B] = A[A, B] + [A, B]A$.

1. Calculating $[\hat{L}_x^2, \hat{L}_z]$: Using the fundamental relation $[\hat{L}_x, \hat{L}_z] = -i\hbar\hat{L}_y$ (derived from $[\hat{L}_z, \hat{L}_x] = i\hbar\hat{L}_y$):

$$\begin{aligned} [\hat{L}_x^2, \hat{L}_z] &= \hat{L}_x[\hat{L}_x, \hat{L}_z] + [\hat{L}_x, \hat{L}_z]\hat{L}_x \\ &= \hat{L}_x(-i\hbar\hat{L}_y) + (-i\hbar\hat{L}_y)\hat{L}_x \end{aligned}$$

$$= -i\hbar(\hat{L}_x\hat{L}_y + \hat{L}_y\hat{L}_x) \quad (93)$$

2. Calculating $[\hat{L}_y^2, \hat{L}_z]$: Using the fundamental relation $[\hat{L}_y, \hat{L}_z] = i\hbar\hat{L}_x$:

$$\begin{aligned} [\hat{L}_y^2, \hat{L}_z] &= \hat{L}_y[\hat{L}_y, \hat{L}_z] + [\hat{L}_y, \hat{L}_z]\hat{L}_y \\ &= \hat{L}_y(i\hbar\hat{L}_x) + (i\hbar\hat{L}_x)\hat{L}_y \\ &= i\hbar(\hat{L}_y\hat{L}_x + \hat{L}_x\hat{L}_y) \end{aligned} \quad (94)$$

Total Commutator: Adding the results:

$$[\hat{L}^2, \hat{L}_z] = -i\hbar(\hat{L}_x\hat{L}_y + \hat{L}_y\hat{L}_x) + i\hbar(\hat{L}_y\hat{L}_x + \hat{L}_x\hat{L}_y) \quad (95)$$

Since scalar multiplication is distributive, the terms cancel exactly:

$$[\hat{L}^2, \hat{L}_z] = 0 \quad (96)$$

By symmetry, it follows that:

$$[\hat{L}^2, \hat{L}_i] = 0, \quad \text{for } i = x, y, z \quad (97)$$

6.2.3 Physical Significance: Simultaneous Eigenstates

The fact that $[\hat{L}^2, \hat{L}_z] = 0$ has profound physical implications in quantum mechanics. Two commuting Hermitian operators share a complete set of simultaneous eigenstates.

This means we can specify the state of the system using the eigenvalues of both operators simultaneously without violating the Heisenberg Uncertainty Principle. Physically, this implies that the magnitude of the angular momentum ($\hat{L}^2$) and its projection along one axis (arbitrarily chosen as the z-axis, $\hat{L}_z$) are compatible observables.

However, since different components do not commute (e.g., $[\hat{L}_x, \hat{L}_y] \neq 0$), we cannot determine all three components simultaneously. Thus, the complete description of the angular momentum state is given by the basis kets $|l, m\rangle$, characterized by the quantum numbers l and m :

- l relates to the eigenvalue of $\hat{L}^2$ ($\hbar^2 l(l+1)$).
- m relates to the eigenvalue of $\hat{L}_z$ ($\hbar m$).

7 Eigenvalue Problem of $\hat{L}^2$ and $\hat{L}_z$

Having established that $[\hat{L}^2, \hat{L}_z] = 0$, we know there exists a common basis of eigenstates for these operators. We define these eigenstates as functions of the spherical angles, denoted by $Y(\theta, \phi)$.

Our goal is to solve the simultaneous eigenvalue equations:

$$\hat{L}_z Y(\theta, \phi) = \lambda Y(\theta, \phi) \quad (98)$$

$$\hat{L}^2 Y(\theta, \phi) = \mu Y(\theta, \phi) \quad (99)$$

where λ and μ are the eigenvalues to be determined.

7.1 Eigenvalue Equation for $\hat{L}_z$

We begin with the simpler operator, $\hat{L}_z$. In spherical coordinates, we derived in Section 4.2.1 that:

$$\hat{L}_z = -i\hbar \frac{\partial}{\partial \phi} \quad (100)$$

The eigenvalue equation is:

$$-i\hbar \frac{\partial}{\partial \phi} Y(\theta, \phi) = \lambda Y(\theta, \phi) \quad (101)$$

7.1.1 Separation of Variables

Since the operator only involves the variable ϕ , we can separate the variables by proposing a solution of the form:

$$Y(\theta, \phi) = f(\theta)\Phi(\phi) \quad (102)$$

Substituting this into the differential equation:

$$-i\hbar f(\theta) \frac{d\Phi(\phi)}{d\phi} = \lambda f(\theta)\Phi(\phi) \quad (103)$$

Assuming $f(\theta) \neq 0$, we can divide by $f(\theta)$ to obtain a first-order ordinary differential equation for $\Phi(\phi)$:

$$\frac{d\Phi}{d\phi} = \frac{i\lambda}{\hbar} \Phi(\phi) \quad (104)$$

7.1.2 General Solution

The solution to this differential equation is an exponential function:

$$\Phi(\phi) = Ae^{i(\lambda/\hbar)\phi} \quad (105)$$

where A is a normalization constant. Consequently, the total eigenfunction takes the form:

$$Y(\theta, \phi) = f(\theta)e^{i(\lambda/\hbar)\phi} \quad (106)$$

7.1.3 Boundary Conditions and Quantization

For the wavefunction to be physically acceptable, it must be single-valued. In spherical coordinates, the angle ϕ and the angle $\phi + 2\pi$ represent the exact same physical point in space. Therefore, the function must satisfy the periodicity condition:

$$\Phi(\phi + 2\pi) = \Phi(\phi) \quad (107)$$

Substituting our solution:

$$e^{i(\lambda/\hbar)(\phi+2\pi)} = e^{i(\lambda/\hbar)\phi} \quad (108)$$

$$e^{i(\lambda/\hbar)\phi} e^{i(\lambda/\hbar)2\pi} = e^{i(\lambda/\hbar)\phi} \quad (109)$$

Dividing both sides by $e^{i(\lambda/\hbar)\phi}$, we require:

$$e^{2\pi i(\lambda/\hbar)} = 1 \quad (110)$$

From Euler's formula ($e^{ix} = \cos x + i \sin x$), this equality holds if and only if the exponent is an integer multiple of $2\pi i$. Let us define a number m such that:

$$\frac{\lambda}{\hbar} = m, \quad \text{where } m \in \mathbb{Z} = \{0, \pm 1, \pm 2, \dots\} \quad (111)$$

7.1.4 Result

This boundary condition restricts the eigenvalues λ to discrete values. We conclude that the eigenvalues of $\hat{L}_z$ are quantized in units of $\hbar$:

$$\lambda = m\hbar \quad (112)$$

And the angular dependence on ϕ is necessarily:

$$Y(\theta, \phi) = f(\theta)e^{im\phi} \quad (113)$$

where m is called the *magnetic quantum number*.

7.2 Eigenvalue Equation for $\hat{L}^2$

We now address the eigenvalue problem for the square of the angular momentum operator. Using the explicit expression for $\hat{L}^2$ in spherical coordinates derived in Eq. (62):

$$\hat{L}^2 = -\hbar^2 \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right] \quad (114)$$

We set up the eigenvalue equation aiming to find the eigenvalue λ :

$$\hat{L}^2 Y(\theta, \phi) = \lambda Y(\theta, \phi) \quad (115)$$

7.2.1 Derivation of the Radial Equation

We substitute the separable solution form obtained in the previous section, $Y(\theta, \phi) = f(\theta)e^{im\phi}$, into the eigenvalue problem.

First, we evaluate the action of the second derivative with respect to ϕ on our trial function:

$$\frac{\partial^2}{\partial \phi^2} (f(\theta)e^{im\phi}) = f(\theta) \frac{\partial^2}{\partial \phi^2} e^{im\phi} = f(\theta)(im)^2 e^{im\phi} = -m^2 f(\theta)e^{im\phi} \quad (116)$$

Substituting this result back into the full operator expression:

$$-\hbar^2 \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial f}{\partial \theta} \right) e^{im\phi} - \frac{m^2}{\sin^2 \theta} f(\theta) e^{im\phi} \right] = \lambda f(\theta) e^{im\phi} \quad (117)$$

Since the phase factor $e^{im\phi}$ is non-zero everywhere, we can divide the entire equation by it. Multiplying by $-1/\hbar^2$, we isolate the differential equation for the polar angle function $f(\theta)$:

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{df}{d\theta} \right) - \frac{m^2}{\sin^2 \theta} f(\theta) = -\frac{\lambda}{\hbar^2} f(\theta) \quad (118)$$

Rearranging all terms to one side, we obtain the differential equation for $f(\theta)$:

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{df}{d\theta} \right) + \left(\frac{\lambda}{\hbar^2} - \frac{m^2}{\sin^2 \theta} \right) f(\theta) = 0 \quad (119)$$

7.3 Natural Emergence of $\hat{L}^2$ and $\hat{L}_z$: Spherical Harmonics

The differential equation derived above is the well-known *Associated Legendre Differential Equation*. To see this clearly, one typically performs the change of variables $x = \cos \theta$, which maps the domain $\theta \in [0, \pi]$ to $x \in [-1, 1]$.

7.3.1 Explicit Substitution

Let us apply the chain rule to transform the derivatives from θ to x . Note that:

$$\frac{d}{d\theta} = -\sin \theta \frac{d}{dx} \quad (120)$$

Substituting this and the identity $\sin^2 \theta = 1 - x^2$ into the eigenvalue equation yields:

$$\frac{d}{dx} \left[(1 - x^2) \frac{df}{dx} \right] + \left(\frac{\lambda}{\hbar^2} - \frac{m^2}{1 - x^2} \right) f(x) = 0 \quad (121)$$

7.3.2 Physical Admissibility: The Origin of Quantization

The differential equation for $f(x)$ contains singular points at the poles $x = \pm 1$ (corresponding to $\theta = 0$ and $\theta = \pi$) due to the term $(1 - x^2)$ in the denominator.

Mathematically, the general solution to the Legendre equation is an infinite power series. Convergence analysis (specifically using the Frobenius method) reveals that this infinite series generally diverges logarithmically at the poles ($x \rightarrow \pm 1$).

Regularity Condition

For the wavefunction to represent a physical state, it must be single-valued, continuous, and finite everywhere (square-integrable). An infinite value at the poles is physically unacceptable. Therefore, we must impose a **regularity condition**: the infinite series must terminate to form a finite polynomial.

This truncation of the series only occurs if the coefficient of the recursion relation vanishes at some index l . This condition strictly demands that the eigenvalue parameter λ satisfies:

$$\frac{\lambda}{\hbar^2} = l(l+1), \quad l \in \mathbb{Z}^+ \quad (122)$$

Thus, the quantization of angular momentum magnitude ($\hat{L}^2$) is not an arbitrary postulate, but a direct consequence of the topological requirement that the wavefunction must be well-behaved on the closed surface of the sphere.

Constraints on m via Rodrigues Formula

With λ fixed by l , the acceptable non-singular solutions are the Associated Legendre Polynomials $P_l^m(x)$. Their construction via the Rodrigues formula clearly elucidates the restriction on the magnetic quantum number m :

$$P_l^m(x) \propto (1-x^2)^{m/2} \frac{d^{l+m}}{dx^{l+m}} [(x^2-1)^l] \quad (123)$$

Notice that the term $(x^2-1)^l$ is a polynomial of degree $2l$.

- If we differentiate a polynomial of degree $2l$ more than $2l$ times, the result is zero.
- However, the structure of the Legendre equation specifically requires the order of the derivative to be $l+m$.
- If $|m| > l$, then $l+m > 2l$ (considering the structure of the differential operator), or more simply, the associated function involves derivatives that would annihilate the polynomial or produce singularities structure incompatible with the equation order.

Specifically, the Rodrigues definition implies that we can only take up to l “vertical” steps (ladder operators) away from the base state. Therefore, for a non-vanishing solution, we must have:

$$|m| \leq l \quad (124)$$

This explains why the projection of angular momentum ($\hat{L}_z$) cannot exceed its total magnitude ($\hat{L}^2$).

7.3.3 Spherical Harmonics Definition

Combining the radial part $P_l^m(\cos \theta)$ with the azimuthal phase $e^{im\phi}$, we obtain the **Spherical Harmonics** $Y_{l,m}(\theta, \phi)$, which form the orthonormal basis on the sphere:

$$Y_{l,m}(\theta, \phi) = (-1)^m \sqrt{\frac{(2l+1)}{4\pi} \frac{(l-m)!}{(l+m)!}} P_l^m(\cos \theta) e^{im\phi} \quad (125)$$

7.4 Simultaneous Eigenstates

We conclude that there exist state vectors, denoted as $|l, m\rangle$, which are simultaneous eigenvectors of $\hat{L}^2$ and $\hat{L}_z$. These states completely describe the angular properties of the free particle (or any central potential system).

The defining eigenvalue equations are:

$$\hat{L}^2 |l, m\rangle = \hbar^2 l(l+1) |l, m\rangle \quad (126)$$

$$\hat{L}_z |l, m\rangle = \hbar m |l, m\rangle \quad (127)$$

These kets $|l, m\rangle$ form the angular momentum representation basis that we will use to solve the Schrödinger equation in the following section.

8 Dynamics and the Radial Wave Equation

We formally solve the Time-Dependent Schrödinger Equation (TDSE) by projecting the dynamics onto the eigenbasis of the Complete Set of Commuting Observables (CSCO) $\{\hat{H}, \hat{L}^2, \hat{L}_z\}$. This procedure separates the trivial temporal evolution from the geometric spatial structure.

8.1 Spectral Evolution

The state vector $|\psi(t)\rangle$ is expanded in the continuous basis of simultaneous eigenstates $|k, \ell, m\rangle$, normalized to the radial measure $k^2 dk$:

$$|\psi(t)\rangle = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} \int_0^{\infty} dk k^2 c_{k\ell m}(t) |k, \ell, m\rangle. \quad (128)$$

Substituting this expansion into the TDSE $i\hbar \partial_t |\psi\rangle = \hat{H} |\psi\rangle$ and exploiting the orthogonality of the basis, the partial differential equation decouples into an infinite set of independent ordinary differential equations for the coefficients $c_{k\ell m}(t)$:

$$i\hbar \dot{c}_{k\ell m}(t) = E_k c_{k\ell m}(t), \quad \text{where } E_k = \frac{\hbar^2 k^2}{2m}. \quad (129)$$

The solution is exact and demonstrates that the angular quantum numbers are constants of motion:

$$c_{k\ell m}(t) = c_{k\ell m}(0) \exp\left(-\frac{i}{\hbar} \frac{\hbar^2 k^2}{2m} t\right). \quad (130)$$

8.2 The Coordinate Representation (Radial Equation)

To determine the spatial probability distribution, we must find the coordinate representation of the basis kets, $\langle \mathbf{r} | k, \ell, m \rangle$. We exploit the separation of variables in spherical coordinates:

$$\langle \mathbf{r} | k, \ell, m \rangle = R_{k\ell}(r) Y_{\ell m}(\theta, \phi). \quad (131)$$

Here, $Y_{\ell m}$ are the Spherical Harmonics derived previously. The radial function $R_{k\ell}(r)$ is determined by the energy eigenvalue equation in position space:

$$\langle \mathbf{r} | \hat{H} | k, \ell, m \rangle = E_k \langle \mathbf{r} | k, \ell, m \rangle. \quad (132)$$

Using the Laplacian decomposition $\nabla^2 = \partial_r^2 + \frac{2}{r}\partial_r - \frac{\hat{L}^2}{\hbar^2 r^2}$, the equation becomes:

$$-\frac{\hbar^2}{2m} \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial R_{k\ell}}{\partial r} \right) - \frac{\ell(\ell+1)}{r^2} R_{k\ell}(r) \right] = \frac{\hbar^2 k^2}{2m} R_{k\ell}(r). \quad (133)$$

8.2.1 The Spherical Bessel Equation

Rearranging terms and introducing the dimensionless variable $\rho = kr$, we obtain the Spherical Bessel differential equation:

$$\frac{d^2 R}{d\rho^2} + \frac{2}{\rho} \frac{dR}{d\rho} + \left[1 - \frac{\ell(\ell+1)}{\rho^2} \right] R(\rho) = 0. \quad (134)$$

The term $\frac{\ell(\ell+1)}{\rho^2}$ represents the centrifugal potential barrier arising from angular momentum conservation, which prevents high- ℓ particles from penetrating near the origin.

8.2.2 Free Spherical Waves

The physically admissible solutions (regular at the origin $r = 0$) are the Spherical Bessel Functions of the first kind, denoted $j_\ell(kr)$.

$$R_{k\ell}(r) = \sqrt{\frac{2}{\pi}} j_\ell(kr). \quad (135)$$

(The normalization factor $\sqrt{2/\pi}$ is chosen to satisfy the delta-function normalization $\int r^2 dr R_{k\ell} R_{k'\ell} = \delta(k - k')/k^2$).

8.3 The Full Wavefunction Solution

Combining the spectral evolution with the coordinate basis, the general solution for the free particle in the angular momentum representation is:

$$\psi(\mathbf{r}, t) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} \int_0^{\infty} dk k^2 \underbrace{c_{k\ell m}(0) e^{-iE_k t/\hbar}}_{\text{Dynamics}} \underbrace{\sqrt{\frac{2}{\pi}} j_\ell(kr) Y_{\ell m}(\theta, \phi)}_{\text{Spatial Mode}}. \quad (136)$$

This expansion represents the free particle as a superposition of spherical standing waves, providing the natural basis for scattering problems where the asymptotic behavior is defined by incoming and outgoing spherical states.

9 Time Evolution and General Solution

The diagonalization of the Hamiltonian within the CSCO basis $\{|k, \ell, m\rangle\}$ decouples the global evolution of the field $\psi(\mathbf{r}, t)$ into an infinite set of independent dynamical modes. We construct the general solution to the initial value problem by applying the unitary time-evolution operator to the spectral expansion.

9.1 Unitary Evolution of Spectral Modes

The equation of motion derived implies that each spectral amplitude $\psi_{k\ell m}(t)$ undergoes simple harmonic oscillation in the complex plane. The formal solution is obtained via the action of the propagator $\hat{U}(t) = \exp(-i\hat{H}t/\hbar)$:

$$\psi_{k\ell m}(t) = \langle k, \ell, m | \hat{U}(t) | \psi(0) \rangle = e^{-iE_k t/\hbar} \psi_{k\ell m}(0). \quad (137)$$

Explicitly, with the free particle dispersion relation $E_k = \frac{\hbar^2 k^2}{2m}$:

$$\psi_{k\ell m}(t) = \psi_{k\ell m}(0) \exp\left(-i \frac{\hbar k^2}{2m} t\right). \quad (138)$$

This result highlights the stationarity of the angular momentum eigenstates; the probability density $|\psi_{k\ell m}(t)|^2$ is time-invariant, ensuring that information encoded in the angular channels is conserved.

9.2 Synthesis of the Dynamical State

The full dynamical state $|\psi(t)\rangle$ is reconstructed by synthesizing the evolved modes. Substituting the spectral solution (138) into the basis expansion yields the general solution in abstract notation:

$$|\psi(t)\rangle = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} \int_0^{\infty} dk k^2 \psi_{k\ell m}(0) e^{-i \frac{\hbar k^2}{2m} t} |k, \ell, m\rangle. \quad (139)$$

Projecting this state onto the coordinate basis $\langle \mathbf{r} |$ and utilizing the free spherical wave functions $\langle \mathbf{r} | k, \ell, m \rangle = \sqrt{\frac{2}{\pi}} j_{\ell}(kr) Y_{\ell m}(\Omega)$, we obtain the explicit position-space wavefunction:

$$\psi(\mathbf{r}, t) = \sqrt{\frac{2}{\pi}} \sum_{\ell, m} Y_{\ell m}(\Omega) \int_0^{\infty} dk k^2 \psi_{k\ell m}(0) j_{\ell}(kr) e^{-i \frac{\hbar k^2}{2m} t}. \quad (140)$$

This integral transform constitutes the complete solution to the free-particle Schrödinger equation in spherical coordinates.

9.3 Physical Interpretation

The structure of Eq. (140) elucidates the physical mechanism of free evolution:

- **Angular Stability:** The quantum numbers ℓ and m appear strictly as summation indices for the stationary harmonics $Y_{\ell m}$. The absence of time dependence in the angular sector confirms that the probability distribution over the orbital angular momentum channels is invariant, a direct consequence of the isotropy of free space ($[\hat{H}, \hat{\mathbf{L}}] = 0$).

- **Radial Dispersion:** Time dependence is confined to the phase modulation of the radial modes. Since the dispersion relation is non-linear ($E_k \propto k^2$), the group velocity $v_g = \partial_k \omega = \hbar k / m$ varies with wavenumber. This gradient in propagation speed across different k -components drives the characteristic wave packet spreading (dispersion) observed in the position representation.

Ultimately, the angular momentum representation resolves the complex diffusive dynamics of the free particle into a static rotation of independent spectral components in the complex plane.

10 Integral Transforms and Representations

The abstract solution $|\psi(t)\rangle$ acquires physical meaning only when projected onto specific basis manifolds. We construct the explicit integral transforms that map the angular momentum spectral amplitudes $\psi_{k\ell m}$ to the observable representations of position, momentum, and energy.

10.1 Position Space: The Spherical Bessel Transform

The transition to coordinate space is defined by the transformation kernel $\langle \mathbf{r} | k, \ell, m \rangle$. This kernel arises from the partial-wave expansion of the plane wave, identifying the free spherical wave as the radial eigenfunction.

10.1.1 The Transformation Kernel

We adopt the normalization convention consistent with the quantum Fourier transform:

$$\langle \mathbf{r} | k, \ell, m \rangle = i^\ell \sqrt{\frac{2}{\pi}} j_\ell(kr) Y_{\ell m}(\Omega_r). \quad (141)$$

The phase factor i^ℓ is essential for time-reversal consistency.

10.1.2 Wavefunction Synthesis

Projecting the spectral expansion Eq. (139) onto the position bra $\langle \mathbf{r} |$ yields the spatial field:

$$\psi(\mathbf{r}, t) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} i^\ell Y_{\ell m}(\Omega_r) \int_0^{\infty} dk k^2 \underbrace{\sqrt{\frac{2}{\pi}} j_\ell(kr)}_{\text{Radial Basis}} \psi_{k\ell m}(t). \quad (142)$$

This expression identifies the radial wavefunction $R_\ell(r, t)$ as the Spherical Bessel Transform (or Hankel transform of order $\ell + 1/2$) of the spectral amplitude $\psi_{k\ell m}$. This is the spherical analog to the standard Fourier transform, mapping the radial momentum domain k to the radial position domain r .

10.2 Momentum Space: Harmonic Analysis on the Sphere

In the linear momentum representation, the physics is simplified by the fact that $\hat{\mathbf{p}}$ commutes with the free Hamiltonian. The angular momentum states define a specific geometric partitioning of momentum space.

10.2.1 Diagonal Representation

The basis states $|k, \ell, m\rangle$ are eigenstates of the momentum magnitude operator $\hat{p} = \sqrt{\hat{\mathbf{p}}^2}$. Consequently, the projection kernel is sharply localized (distributional):

$$\langle \mathbf{p} | k, \ell, m \rangle = \frac{\delta(p - \hbar k)}{p^2} Y_{\ell m}(\Omega_p). \quad (143)$$

Here, Ω_p denotes the angles of the momentum vector.

10.2.2 Momentum Distribution

Substituting this kernel into the general solution collapses the radial integral immediately:

$$\begin{aligned} \psi(\mathbf{p}, t) &= \int_0^\infty dk k^2 \psi_{k\ell m}(t) \left[\frac{\delta(p - \hbar k)}{p^2} Y_{\ell m}(\Omega_p) \right] \\ &= \frac{1}{\hbar^3} \psi_{p/\hbar, \ell, m}(t) Y_{\ell m}(\Omega_p). \end{aligned} \quad (144)$$

This result confirms that the spectral coefficients $\psi_{k\ell m}$ are physically equivalent to the momentum-space wavefunction projected onto spherical harmonics. The time evolution is strictly a phase rotation, as momentum states are stationary solutions of the free particle.

10.3 Energy Representation and Density of States

The energy representation is crucial for scattering theory (S-matrix formalism). Since the spectrum of the free particle is continuous and degenerate, the mapping involves the Density of States (DOS).

10.3.1 Jacobian Transformation

Using the dispersion relation $E = \frac{\hbar^2 k^2}{2m}$, we relate the differential measures:

$$dk = \frac{dk}{dE} dE = \frac{1}{\hbar} \sqrt{\frac{m}{2E}} dE. \quad (145)$$

The density of states factor is $\rho(E) \propto \sqrt{E}$.

10.3.2 Spectral Evolution

The wavefunction in the energy representation $\psi(E, \ell, m)$ is related to the k -space amplitude by preserving the probability norm $\int |\psi|^2 dk = \int |\phi|^2 dE$:

$$\psi(E, \ell, m; t) = \left(\frac{mk(E)}{\hbar^2} \right)^{1/2} \psi_{k(E), \ell, m}(0) e^{-iEt/\hbar}. \quad (146)$$

This representation trivializes the dynamics to a linear phase evolution, explicitly demonstrating that the angular momentum modes are the stationary states of the system.

11 Synthesis of the Non-Relativistic Formalism

The construction of the angular momentum representation for the free particle serves as a complete laboratory for the axioms of non-relativistic quantum mechanics. We conclude by abstracting the mathematical structure into a set of formal postulates and defining the geometric decomposition of the state space.

11.1 Fundamental Postulates

Based on the spectral analysis of the operators $\hat{\mathbf{r}}$, $\hat{\mathbf{p}}$, $\hat{\mathbf{L}}$, and $\hat{H}$, we encapsulate the theory in four structural definitions:

1. **The State Space:** The physical state of the system is represented by a ray in a complex, separable Hilbert space $\mathcal{H} \cong L^2(\mathbb{R}^3)$.
2. **Observables:** Measurable physical quantities correspond to self-adjoint (Hermitian) operators acting on $\mathcal{H}$. The possible outcomes of a measurement are restricted to the spectrum $\sigma(\hat{A})$ of the operator.
3. **Born Rule (Spectral Projection):** For a system in state $|\psi\rangle$, the probability of measuring an eigenvalue $\lambda \in \sigma(\hat{A})$ is given by the expectation value of the projection operator $\hat{P}_\lambda$:

$$P(\lambda) = \langle \psi | \hat{P}_\lambda | \psi \rangle = |\langle \lambda | \psi \rangle|^2. \quad (147)$$

4. **Unitary Evolution:** The time evolution of a closed system is deterministic and generated by the Hamiltonian $\hat{H}$, forming a strongly continuous unitary group $\hat{U}(t) = e^{-i\hat{H}t/\hbar}$.

11.2 Hilbert Space Decomposition and the CSCO

The rotational symmetry of the free Hamiltonian implies a specific geometric structure for the Hilbert space. We utilize the Complete Set of Commuting Observables (CSCO) $\mathcal{S} = \{\hat{H}, \hat{L}^2, \hat{L}_z\}$ to partition the space.

11.2.1 Tensor Product Structure

The Hilbert space factorizes into a radial component and an angular component:

$$\mathcal{H} \cong \mathcal{H}_{\text{radial}} \otimes \mathcal{H}_{\text{angular}} = L^2(\mathbb{R}^+, r^2 dr) \otimes L^2(S^2, d\Omega). \quad (148)$$

11.2.2 Invariant Subspaces

Under the action of the rotation group $\text{SO}(3)$, the angular space decomposes into a direct sum of orthogonal invariant subspaces $\mathcal{E}^{(\ell)}$, each of dimension $2\ell + 1$. The total Hilbert space is the direct sum:

$$\mathcal{H} = \bigoplus_{\ell=0}^{\infty} \mathcal{H}^{(\ell)}. \quad (149)$$

Each subspace $\mathcal{H}^{(\ell)}$ contains states with definite total angular momentum ℓ , which remain invariant in magnitude under rotations, mixing only among the m -components.

11.2.3 The Symmetry-Adapted Eigenbasis

The basis vectors spanning these subspaces are the simultaneous eigenvectors of the CSCO, denoted $|E, \ell, m\rangle$ (or $|k, \ell, m\rangle$):

$$\hat{H} |k, \ell, m\rangle = \frac{\hbar^2 k^2}{2m} |k, \ell, m\rangle, \quad (150)$$

$$\hat{L}^2 |k, \ell, m\rangle = \hbar^2 \ell(\ell + 1) |k, \ell, m\rangle, \quad (151)$$

$$\hat{L}_z |k, \ell, m\rangle = \hbar m |k, \ell, m\rangle. \quad (152)$$

This basis diagonalizes the dynamics, reducing the 3D Schrödinger PDE to a 1D radial ODE for each ℓ -channel. The Completeness Relation (Resolution of Identity) for this decomposition is:

$$\hat{\mathbb{I}} = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} \int_0^{\infty} dk |k, \ell, m\rangle \langle k, \ell, m|. \quad (153)$$

This summation explicitly reconstructs the full space from the irreducible representations of the symmetry group.

11.3 Symmetries and Conservation Laws (Quantum Noether's Theorem)

The derivation of the angular momentum eigenvalues exhibits the profound link between continuous symmetries and conservation laws. In the quantum regime, Noether's theorem is realized through the algebraic properties of the unitary generators of symmetry groups.

11.3.1 Invariance and Generators

Let $\mathcal{G}$ be a continuous group of transformations acting on the Hilbert space, parameterized by a real variable α and generated by the self-adjoint operator $\hat{G}$:

$$\hat{U}(\alpha) = \exp\left(-\frac{i}{\hbar}\alpha\hat{G}\right). \quad (154)$$

The system is invariant under this symmetry if the Hamiltonian commutes with the unitary transformation, $[\hat{H}, \hat{U}(\alpha)] = 0$, which for infinitesimal transformations implies:

$$[\hat{H}, \hat{G}] = 0. \quad (155)$$

By the Heisenberg equation of motion, this commutation ensures that the generator $\hat{G}$ is a constant of motion:

$$\frac{d}{dt}\hat{G}_H(t) = \frac{1}{i\hbar}[\hat{G}_H, \hat{H}] = 0. \quad (156)$$

11.3.2 Symmetries of the Free Particle

For the free particle Hamiltonian $\hat{H} = \hat{\mathbf{p}}^2/2m$, the maximal symmetry group of Euclidean space dictates the fundamental conservation laws:

- **Spatial Homogeneity (Translation Group T_3):** Invariant under $\mathbf{r} \rightarrow \mathbf{r} + \mathbf{a}$.

$$[\hat{H}, \hat{\mathbf{p}}] = 0 \implies \frac{d\langle\hat{\mathbf{p}}\rangle}{dt} = 0 \quad (\text{Linear Momentum}). \quad (157)$$

- **Spatial Isotropy (Rotation Group $SO(3)$):** Invariant under $\mathbf{r} \rightarrow \mathcal{R}\mathbf{r}$.

$$[\hat{H}, \hat{\mathbf{L}}] = 0 \implies \frac{d\langle\hat{\mathbf{L}}\rangle}{dt} = 0 \quad (\text{Angular Momentum}). \quad (158)$$

- **Time Homogeneity:** Invariant under $t \rightarrow t + \tau$.

$$[\hat{H}, \hat{H}] = 0 \implies \frac{d\langle\hat{H}\rangle}{dt} = 0 \quad (\text{Energy}). \quad (159)$$

11.3.3 Classical Correspondence: The Poisson Bracket

The structural unity between classical and quantum mechanics is established via the Dirac quantization rule. Classically, the time evolution of an observable A is governed by the Poisson bracket with the Hamiltonian:

$$\frac{dA}{dt} = \{A, H\}_{PB} + \frac{\partial A}{\partial t}. \quad (160)$$

The quantum commutator is the algebraic isomer of the Poisson bracket, scaled by the deformation parameter $\hbar$. In the limit $\hbar \rightarrow 0$, we recover the classical law:

$$\lim_{\hbar \rightarrow 0} \frac{1}{i\hbar} [\hat{A}, \hat{H}] \longrightarrow \{A, H\}_{PB}. \quad (161)$$

Consequently, the quantum statement $[\hat{L}_i, \hat{H}] = 0$ corresponds directly to the classical statement $\{L_i, H\}_{PB} = 0$. This isomorphism ensures that the conservation of angular momentum is a robust feature of the central force problem, valid across both microscopic and macroscopic regimes.

11.4 Classical-Quantum Isomorphism and Formalism

The structural correspondence between the classical symplectic manifold and the quantum Hilbert space is preserved via the Dirac quantization map. This mapping is not merely a syntactic substitution but represents a functorial transition between two distinct mathematical categories.

11.4.1 The Dynamical Dictionary

The following dictionary establishes the algebraic isomorphism between the commutative algebra of classical functions and the non-commutative algebra of quantum operators:

Table 1 Structural Correspondence: Geometry vs. Algebra

Concept	Classical Mechanics (Geometric)	Quantum Mechanics (Algebraic)
Arena	Symplectic Manifold $\mathcal{M} \cong T^*\mathbb{Q}$	Complex Hilbert Space $\mathcal{H}$
State	Point $\xi = (\mathbf{r}, \mathbf{p}) \in \mathcal{M}$	Ray $ \psi\rangle \in \mathbb{P}\mathcal{H}$
Observables	Real Functions $C^\infty(\mathcal{M})$	Hermitian Operators $\mathcal{B}_{sa}(\mathcal{H})$
Structure	Commutative, Associative	Non-Commutative, Associative
Lie Bracket	Poisson Bracket $\{A, B\}_{PB}$	Commutator $\frac{1}{i\hbar} [\hat{A}, \hat{B}]$
Generator	Hamiltonian Vector Field X_H	Hamiltonian Operator $\hat{H}$
Dynamics	$\frac{df}{dt} = \{f, H\}_{PB}$	$\frac{d\hat{A}}{dt} = \frac{1}{i\hbar} [\hat{A}, \hat{H}]$

11.4.2 From Differential Geometry to Algebraic Topology

This dictionary highlights a fundamental shift in mathematical formalism:

- **Classical Mechanics as Differential Geometry:** Classical dynamics is the study of Differential Topology and Symplectic Geometry. The state is a point on a smooth differentiable manifold. Time evolution is a continuous flow generated by a vector field tangent to the manifold. The algebra of observables is commutative (order of multiplication does not matter), reflecting the geometric commutativity of coordinate functions.

- **Quantum Mechanics as Algebraic Geometry:** Quantum mechanics abandons the point-set topology of the manifold in favor of Non-Commutative Geometry (Algebraic Topology). The uncertainty principle $[\hat{x}, \hat{p}] \neq 0$ eliminates the concept of a phase space “point”. Instead, the theory is defined by the algebraic structure of the operators themselves (C^* -algebra). The geometry is “spectral”, encoded in the eigenvalues and eigenstates of the algebra rather than spatial coordinates.

11.5 The Unifying Role of Clifford Algebras

While the formalisms appear distinct, Clifford Algebra (Geometric Algebra) provides the bridge that elucidates the geometric origin of algebraic quantum structures.

1. **The Bivector Nature of Rotation:** In both classical and quantum mechanics, angular momentum is fundamentally a bivector quantity, $\mathbf{L} \sim \mathbf{r} \wedge \mathbf{p}$.
 - In the differential geometry of CM, this bivector generates flows along closed curves on the manifold.
 - In the algebraic geometry of QM, this bivector appears in the commutation relations of the generators of the Lie algebra $\mathfrak{so}(3)$.
2. **Spin and Topology:** Differential geometry on $SO(3)$ captures integer angular momentum ($\ell \in \mathbb{Z}$). However, the algebraic structure of Clifford algebras naturally reveals the Spinor representations (half-integer spin) which arise from the simply connected covering group $SU(2) \cong \text{Spin}(3)$.

Thus, Clifford algebras demonstrate that the “algebraic” quirks of quantum mechanics (like non-commutativity and spin) are actually deep “geometric” properties of spacetime that classical differential geometry fails to resolve.

12 Relativistic Quantum Mechanics and the Origin of Spin

12.1 The Poincaré Group and Particle Representations

The unification of special relativity with quantum mechanics necessitates that physical states transform consistently under the symmetries of spacetime. The full isometry group of Minkowski spacetime is the Poincaré group, $IO(1, 3) \cong T(1, 3) \rtimes O(1, 3)$, which combines spacetime translations with Lorentz transformations. Within the Wigner-Bargmann framework, elementary particles are defined as irreducible unitary representations of this group.

While the Klein-Gordon equation corresponds to the scalar representation (spin-0), carrying the trivial representation of the Lorentz group, this is insufficient to describe matter particles (fermions). Fermions require a projective (spinor) representation, specifically the $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$ representation, to account for their intrinsic angular momentum. Consequently, a deep analysis of relativistic wave equations is equivalent to studying the fundamental representation theory of spacetime symmetries.

12.2 Derivation of the Dirac Equation

The derivation begins with the relativistic Hamiltonian for a free particle:

$$H = \sqrt{p^2 c^2 + m^2 c^4}. \quad (162)$$

Direct application of the correspondence principle ($E \rightarrow i\hbar\partial_t$, $p \rightarrow -i\hbar\nabla$) to this Hamiltonian results in a wave equation containing a square-root operator, which acts as a non-local pseudo-differential operator. To avoid this, one typically squares the energy relation to $E^2 = p^2 c^2 + m^2 c^4$, yielding the Klein-Gordon equation. However, this second-order equation leads to non-positive definite probability densities and fails to describe spin-1/2 particles.

To resolve these issues, Dirac insisted on an equation that was first-order in the time derivative (like Schrödinger's) to ensure a positive-definite probability density. Relativistic covariance then demands that the equation must also be first-order in spatial derivatives. Dirac proposed a first-order Lorentz-covariant operator $D = \gamma_\mu \partial_\mu$ that “squares” to the d'Alembertian operator of the Klein-Gordon equation ($D^2 = c^{-2}\partial_t^2 - \nabla^2$). This requirement leads to the Dirac equation:

$$i\hbar \left(\gamma_0 \frac{1}{c} \frac{\partial}{\partial t} + \gamma_1 \frac{\partial}{\partial x_1} + \gamma_2 \frac{\partial}{\partial x_2} + \gamma_3 \frac{\partial}{\partial x_3} \right) \psi = mc\psi. \quad (163)$$

12.3 Encoding Spin and Antimatter

A relativistic first-order equation is impossible if the coefficients are scalars. For the operator D to factorize the Laplacian, the coefficients γ_μ must satisfy the defining anticommutation relations of the Clifford algebra $\mathcal{Cl}_{1,3}(\mathbb{R})$:

$$\{\gamma_\mu, \gamma_\nu\} = \gamma_\mu \gamma_\nu + \gamma_\nu \gamma_\mu = 2\eta_{\mu\nu} \mathbb{I}. \quad (164)$$

This algebraic structure dictates that the wavefunction ψ cannot be a scalar but must be a spinor residing in a representation space of the algebra. Thus, spin is not an ad hoc addition but an automatic consequence of linearizing the relativistic energy relation; the equation naturally encodes the magnetic moment with gyromagnetic ratio $g = 2$. Furthermore, this structure necessitates the existence of charge-conjugate solutions, predicting the existence of antimatter (the positron).

12.4 Bosons, Fermions, and Statistics

The distinction between bosons and fermions arises from their behavior under rotations. Lorentz transformations are realized as inner automorphisms of the Clifford algebra acting via spin groups, which provide a covering of the $SO^+(3,1)$ group. While bosonic fields (like those obeying the Klein-Gordon equation) transform under the standard vector or scalar representations ($SO(3)$), fermionic fields transform under spinor representations ($SU(2)$), acquiring a phase factor of -1 under a 2π rotation.

This geometric distinction induces the statistical nature of the particles. Understanding the Clifford algebra is essential to grasp the distinction between the Commutative

algebras governing Bosonic fields and the Grassmann-graded algebras required for Fermionic matter. These anticommutation relations constitute the finite-dimensional archetype of the Canonical Anticommutation Relations (CAR) in Quantum Field Theory, leading directly to the antisymmetry of multiparticle wavefunctions and the Pauli exclusion principle.

References